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**Ramesh_Cisco.pdf**
PDF

what technical interview question will dristee foundation asks, 30 min interview, Find ALL online sources for this, give it in proper q/a format, i am attaching my resume,

I went through your resume carefully (strong profile btw: backend + systems + low-latency work + DSA). For a **Dristee Foundation–type 30 min technical interview**, expect **focused, practical + fundamental questions**, not FAANG-level puzzles but **clear thinking, fundamentals, and project depth**.

There's **no huge public question bank specific to Dristee**, but based on:

- similar NGO/education-tech hiring patterns
- your profile (backend + systems + DSA)
- typical 30-min screening structure

I'll give you a **realistic, high-probability question set in proper Q/A format**.

# 🔥 30-Min Interview Structure (what will actually happen)

- **5 min → Resume discussion**
- **10–15 min → Core CS (DSA/DBMS/OS)**
- **10 min → Project deep dive**
- **5 min → Quick coding or logic**

# 🧠 SECTION 1: Resume-Based Questions (HIGH PRIORITY)

## Q1. Explain your TradeStream Engine in simple terms.

**Answer:**

It is a low-latency crypto order execution engine that:

- connects via WebSockets to exchanges (Deribit)
- processes real-time order book data
- executes trades with minimal delay (~334ms from 2.5s)

Key optimizations:

- lock-free data structures

- efficient message parsing
- async networking

## Q2. What are lock-free data structures? Why did you use them?

**Answer:**

- They allow multiple threads to operate **without mutex locks**
- Use atomic operations (CAS)

**Why:**

- Avoid thread blocking
- Reduce latency → critical in trading systems

## Q3. What is WebSocket vs REST API?

**Answer:**

| Feature | REST | WebSocket |
|---|---|---|

| Feature | REST | WebSocket |
|---|---|---|
| Communication | Request-response | Full-duplex |
| Real-time | ❌ | ✅ |
| Overhead | Higher | Lower |

Used WebSockets because:

- real-time price updates required

## Q4. How did you reduce latency from 2.5s to 334ms?

**Answer:**

- eliminated blocking calls
- optimized parsing
- used persistent connections
- improved data structures

## Q5. Explain your ERP backend project.

**Answer:**

- built using PHP + SQL
- handled:
  - authentication
  - role-based access
  - validation

Focused on:

- secure queries
- transaction handling

# 🧮 SECTION 2: DSA Questions (VERY LIKELY)

## Q6. What is the difference between stack and queue?

**Answer:**

- Stack → LIFO
- Queue → FIFO

## Q7. Find duplicate in array (basic coding)

**Answer (Python):**

Python Run

```python
def find_duplicate(arr):
    seen = set()
    for x in arr:
        if x in seen:
            return x
        seen.add(x)
```

## Q8. What is time complexity of binary search?

**Answer:**

- O(log n)

## Q9. Reverse a linked list

**Answer:**

Python Run

```
def reverse(head):
    prev = None
    curr = head
    while curr:
        nxt = curr.next
        curr.next = prev
        prev = curr
        curr = nxt
    return prev
```

## Q10. What is hashing?

**Answer:**

- Mapping data → fixed-size value
- Used in hash tables for O(1) lookup

# 🗄️ SECTION 3: DBMS (VERY COMMON)

## Q11. What is indexing?

**Answer:**

- Improves query speed

- works like a book index

## Q12. Difference: SQL vs NoSQL

**Answer:**

| SQL | NoSQL |
|---|---|
| Structured | Flexible |
| ACID | BASE |
| Tables | Documents/Key-value |

## Q13. What is normalization?

**Answer:**

- removing redundancy
- improving consistency

## Q14. What is a transaction?

**Answer:**

- group of operations executed as one unit
- follows ACID properties

## Q15. What is ACID?

**Answer:**

- Atomicity
- Consistency
- Isolation
- Durability

# ⚙️ SECTION 4: OS / SYSTEMS

## Q16. What is a process vs thread?

**Answer:**

- Process → independent memory
- Thread → shared memory

### Q17. What is deadlock?

**Answer:**

- processes waiting forever for resources

### Q18. What is context switching?

**Answer:**

- switching CPU from one process to another

## 🌐 SECTION 5: Networking (IMPORTANT for YOU)

### Q19. What is TCP vs UDP?

**Answer:**

| TCP | UDP |
|---|---|
| Reliable | Fast |

| TCP | UDP |
| --- | --- |
| Ordered | Unordered |

## Q20. What happens when you type a URL?

**Answer:**

1. DNS lookup
2. TCP handshake
3. HTTP request
4. Server response

# 💡 SECTION 6: Practical Thinking

## Q21. How would you design a chat system?

**Answer:**

- WebSockets
- message queue
- database for storage

### Q22. How would you handle 1000+ users?

**Answer:**

- load balancing
- caching
- async processing

## ⚡ SECTION 7: Behavioral (Short)

### Q23. Tell me about a challenge.

→ Talk about:

- latency optimization in TradeStream

### Q24. Why do you want to work here?

**Answer:**

- impact-driven organization

- opportunity to apply tech for real-world problems

## 🎯 MOST IMPORTANT FOR YOU

Focus heavily on:

- WebSockets
- Low latency systems
- SQL queries
- DSA basics (not advanced)

## 🚨 Reality Check (Important)

They will NOT:

- ask crazy competitive programming questions
- expect deep system design

They WILL:

- test clarity
- test fundamentals
- test if you actually built your projects

## ✅ Final Strategy (Guaranteed Impact)

If you do ONLY this, you'll stand out:

1. Prepare **TradeStream deeply**
2. Revise:
    - SQL queries
    - linked list
    - hashing
3. Be ready to explain:
    - WebSockets
    - latency optimization

If you want next level prep, I can:

- simulate a **real 30-min mock interview**
- ask you questions like interviewer
- give feedback on your answers

Just say: **"take my mock interview"** 🚀

Q1. Please share your work Experience Q2. What are the relavent experiences Q3. Do you comfortable for Field visit. Interview
Conversational type where you are asked to brief about your background and experience and questions regarding your experience are asked along. Your background to fit for the role is not necessary. Q1. Since you have SEO knowledge how would use that skill as a Vidushi Fellow? Q2. Why do you want to work at Drishtee? i could find only these much summarise it and give it in proper q/a Format

Got it — I'll clean this up into a **proper interview-ready Q/A format**, and also **upgrade the answers based on your resume + role expectations (Vidushi Fellow / Drishtee-type org)** so you don't sound generic.

## 🎯 Drishtee / Vidushi Fellow Interview — Final Q&A Set

## 🧑‍💼 Q1. Please share your work experience.

**Answer:**
I am currently a Computer Science undergraduate with hands-on experience in backend development and real-time systems.

- During my internship at **Spacey Science**, I developed a scalable progress tracking module using Node.js and React, improving efficiency and reducing manual work.
- At my college internship, I worked on building a **College ERP backend** using PHP and SQL, focusing on authentication, validation, and database design.
- I also worked on real-time systems at GoQuant, where I integrated APIs for market data streaming and order execution.

Additionally, I have built projects like a **low-latency trading engine** and a **full-stack chess application**, which strengthened my problem-solving and system design skills.

## 💼 Q2. What are your relevant experiences for this role?

**Answer:**
Although my background is technical, the relevant part for this role is:

- **Problem-solving ability** – I've solved 800+ DSA problems, which helps in structured thinking.

- **System building experience** – I've worked on scalable applications used by real users.
- **Adaptability** – I've worked across different tech stacks and environments.
- **Communication through projects** – I've built user-facing systems, so I understand usability and impact.

I believe these skills transfer well into solving real-world problems in communities.

## 🚶 Q3. Are you comfortable with field visits?

**Answer:**
Yes, I am comfortable with field visits.

I understand that roles like this require direct interaction with communities, understanding real problems on the ground, and adapting solutions accordingly. I see field exposure as an opportunity to learn beyond technical knowledge and create meaningful impact.

## 💬 Q4. (Conversational) Tell me about yourself.

**Answer:**
I'm a Computer Science student passionate about building efficient and scalable systems.

I enjoy solving problems—both algorithmic and real-world. I've worked on backend systems, real-time applications, and low-latency systems, and I'm particularly interested in applying technology to practical use cases.

Beyond technical skills, I'm curious and adaptable, and I like understanding how systems impact users in real life.

## 🔍 Q5. Since you have SEO knowledge, how would you use that skill as a Vidushi Fellow?

**Answer (IMPORTANT — this is where you stand out):**
I would use SEO not just for visibility, but for **impact amplification**.

For example:

- Improving online presence of rural entrepreneurs or SHGs by optimizing their listings or content
- Creating simple, search-friendly digital content so their products/services reach a wider audience
- Helping local initiatives appear in search results when people look for relevant services

In short, I would use SEO as a tool to **bridge the gap between grassroots initiatives and digital reach**.

### ❤️ Q6. Why do you want to work at Drishtee?

**Answer (Make this human, not corporate):**
I'm interested in working at Drishtee because it focuses on creating real impact at the grassroots level.

Most of my work so far has been technical, but I want to apply my skills in a way that directly improves people's lives. Drishtee's approach of empowering rural communities aligns with that goal.

I also see this as an opportunity to step out of a purely technical environment and gain a deeper understanding of real-world challenges.

## ⚠️ Important Insight (Don't Miss This)

This interview is **NOT technical-heavy**.

They are testing:

- clarity of thought
- willingness to work in field
- ability to connect skills to impact

## 🚀 Pro Tip (Game Changer)

When answering:

👉 Always connect **tech** → **real-world impact**

Example:

> Not just "I know backend"
> Say: "I can build systems that help scale real-world solutions"

If you want, I can:

- simulate **this exact interview live**
- correct your answers line-by-line
- give you **perfect speaking scripts (natural, not robotic)**

Just say: **"mock me for Drishtee"** 🎯